Lehrstuhl für Regelungstechnik
TUM School of Engineering and Design
Technische Universität München

Semester Thesis:
Moving Horizon Estimation of Tire Friction for MPC in Autonomous Racing

Introduction: Autonomous racing is a growing field of motorsports. Autonomous driving competitions not only offer an inherent entertainment value but are also an extreme testing environment for developing autonomous driving systems. As part of the TUM Autonomous Motorsports Team, the automotive group at the Chair of Automatic Control develops planning and control algorithms for racing in series, such as the Indy Autonomous Challenge (IAC) and the Abu Dhabi Autonomous Racing League (A2RL).



Problem description: In the control module, a Model Predictive Control (MPC) algorithm calculates the optimal steering, throttle and brake inputs based on a vehicle model. The model describes the tire forces with the Magic Formula, a curve of tire force over slip angle. As the tires heat up and wear, the peak of this curve changes in both height and position. A friction scaling factor ($\lambda_{\mu y}$) sets the height of the peak, and the stiffness factor (B) stretches the curve and so sets the slip angle of the peak. If the controller assumes too much grip, the car slides; if too little, it loses lap time.

Recursive least squares (RLS) is a simple online estimator, but it needs parameters that enter the model linearly. This holds for the friction scaling but not for the stiffness factor, so RLS can only approximate a changing peak by scaling its height. RLS also takes the slip angles from the state estimator (an extended Kalman filter), and near the friction limit small errors in these states bias the estimate. Moving Horizon Estimation (MHE) estimates the states and both tire parameters together over a window of past measurements, which avoids the bias and captures the peak position at a higher computational cost.

The thesis is a scientific investigation and comparison of both approaches, not a direct performance improvement of the controller. The goal is to find out when each estimator works better and how the estimates affect the MPC in closed loop. Any result is valid.

Tasks:

- Literature review in tire friction estimation and moving horizon estimation
- Preliminary tests of the estimators in the existing Python acados simulation
- Step-wise comparison: RLS and MHE of the friction scaling, then MHE of both parameters
- Evaluation in the double-track vehicle simulator, including closed-loop tests with the MPC

Prerequisites:

- Solid background in control engineering, state estimation (Kalman filter) and optimization
- Experience in software development with Python (C++ is a plus)
- Interest in vehicle technology and autonomous driving systems
- Analytical, problem-solving mindset and a high degree of autonomy.

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